

Section 5.2 – 5.3 Exercise Answers

1. Quadrant 3
2. Quadrant 1
3. Quadrant 2
4. Quadrant 4
5. 1
6. $\frac{\sqrt{3}}{2}$
7. 0
8. $\frac{1}{2}$
9. $\frac{\sqrt{2}}{2}$
10. $\frac{\sqrt{2}}{2}$
11. $\frac{1}{2}$
12. 0
13. -1
14. -1
15. 1
16. $\frac{\sqrt{3}}{2}$
17. 0
18. 60°
19. 10°
20. 80°
21. 45°
22. 45°
23. $\frac{\pi}{4}$
24. $\frac{\pi}{3}$
25. $\frac{\pi}{6}$
26. $\frac{\pi}{3}$
27. $\frac{\pi}{4}$
28. $\frac{\pi}{8}$
29. 45° ; Quadrant 3; $\sin(225^\circ) = -\frac{\sqrt{2}}{2}$; $\cos(225^\circ) = -\frac{\sqrt{2}}{2}$
30. 60° ; Quadrant 4; $\sin(300^\circ) = -\frac{\sqrt{3}}{2}$; $\cos(300^\circ) = \frac{1}{2}$
31. 45° ; Quadrant 2; $\sin(135^\circ) = \frac{\sqrt{2}}{2}$; $\cos(135^\circ) = -\frac{\sqrt{2}}{2}$
32. 30° ; Quadrant 3; $\sin(210^\circ) = -\frac{1}{2}$; $\cos(210^\circ) = -\frac{\sqrt{3}}{2}$
33. 60° ; Quadrant 2; $\sin(120^\circ) = \frac{\sqrt{3}}{2}$; $\cos(120^\circ) = -\frac{1}{2}$
34. 30° ; Quadrant 2; $\sin(150^\circ) = \frac{1}{2}$; $\cos(150^\circ) = -\frac{\sqrt{3}}{2}$
35. $\frac{\pi}{4}$; Quadrant 3; $\sin\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2}$; $\cos\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2}$
36. $\frac{\pi}{6}$; Quadrant 3; $\sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$; $\cos\left(\frac{7\pi}{6}\right) = -\frac{\sqrt{3}}{2}$

37. $\frac{\pi}{3}$; Quadrant 4; $\sin\left(\frac{5\pi}{3}\right) = -\frac{\sqrt{3}}{2}$; $\cos\left(\frac{5\pi}{3}\right) = \frac{1}{2}$
38. $\frac{\pi}{4}$; Quadrant 2; $\sin\left(\frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2}$; $\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$
39. $\frac{\pi}{3}$; Quadrant 3; $\sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}$; $\cos\left(\frac{4\pi}{3}\right) = -\frac{1}{2}$
40. $\frac{\pi}{3}$; Quadrant 2; $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$; $\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$
41. $\frac{\pi}{6}$; Quadrant 2; $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$; $\cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$
42. $\frac{\pi}{4}$; Quadrant 4; $\sin\left(\frac{7\pi}{4}\right) = -\frac{\sqrt{2}}{2}$; $\cos\left(\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2}$
43. $-\frac{4\sqrt{3}}{7}$
44. $\frac{\sqrt{77}}{9}$
45. $-\frac{\sqrt{55}}{8}$
46. $-\frac{\sqrt{15}}{4}$
47. $(-\infty, \infty)$
48. $[-1, 1]$
49. $\sin t = \frac{\sqrt{2}}{2}$; $\cos t = \frac{\sqrt{2}}{2}$
50. $\sin t = \frac{1}{2}$; $\cos t = -\frac{\sqrt{3}}{2}$
51. $\sin t = -\frac{\sqrt{3}}{2}$; $\cos t = \frac{1}{2}$
52. $\sin t = -\frac{\sqrt{2}}{2}$; $\cos t = -\frac{\sqrt{2}}{2}$
53. $\sin t = \frac{\sqrt{3}}{2}$; $\cos t = \frac{1}{2}$
54. $\sin t = \frac{\sqrt{3}}{2}$; $\cos t = -\frac{1}{2}$
55. $\sin t = -\frac{\sqrt{3}}{2}$; $\cos t = -\frac{1}{2}$
56. $\sin t = -\frac{\sqrt{2}}{2}$; $\cos t = \frac{\sqrt{2}}{2}$
57. $\sin t = 0$; $\cos t = 1$
58. $\sin t = 0$; $\cos t = -1$
59. $\sin t = \frac{\sqrt{2}}{2}$; $\cos t = -\frac{\sqrt{2}}{2}$
60. $\sin t = \frac{1}{2}$; $\cos t = \frac{\sqrt{3}}{2}$
61. $\sin t = -\frac{1}{2}$; $\cos t = -\frac{\sqrt{3}}{2}$
62. $\sin t = -\frac{1}{2}$; $\cos t = \frac{\sqrt{3}}{2}$
63. $\sin t = -1$; $\cos t = 0$
64. $\sin t = 1$; $\cos t = 0$
65. $\frac{\sqrt{3}}{3}$
66. $\frac{2\sqrt{3}}{3}$
67. 2
68. $\sqrt{3}$
69. 1
70. $\sqrt{2}$

71. $\sqrt{2}$

72. 1

73. $\sqrt{3}$

74. 2

75. $\frac{2\sqrt{3}}{3}$

76. $\frac{\sqrt{3}}{3}$

77. $-\frac{\sqrt{3}}{3}$

78. $-\frac{2\sqrt{3}}{3}$

79. -2

80. $\sqrt{3}$

81. -1

82. $-\sqrt{2}$

83. $-\sqrt{2}$

84. -1

85. $-\sqrt{3}$

86. -2

87. $\frac{2\sqrt{3}}{3}$

88. $-\frac{\sqrt{3}}{3}$

89. 1

90. 2

91. 2

92. $\frac{\sqrt{3}}{3}$

93. $-\frac{\sqrt{3}}{3}$

94. -2

95. -2

96. -1

97. $\cos t = -\frac{\sqrt{7}}{4}, \sec t = -\frac{4\sqrt{7}}{7}, \csc t = \frac{4}{3}, \tan t = \frac{3\sqrt{7}}{7}, \cot t = -\frac{\sqrt{7}}{3}$

98. $\sin t = -\frac{2\sqrt{2}}{3}, \sec t = -3, \csc t = \frac{-3\sqrt{2}}{4}, \tan t = 2\sqrt{2}, \cot t = \frac{\sqrt{2}}{4}$

99. $\sin t = \frac{12}{13}, \cos t = \frac{5}{13}, \sec t = \frac{13}{5}, \csc t = \frac{13}{12}, \cot t = \frac{5}{12}$

100. $\sec t = 2, \csc t = \frac{2\sqrt{3}}{3}, \tan t = \sqrt{3}, \cot t = \frac{\sqrt{3}}{3}$

101. $-\frac{\sqrt{2}}{2}$

102. $\frac{1}{2}$

103. 3.1

104. -0.34

105. 1.4

106. -9.23

107. $\sin t = \frac{\sqrt{2}}{2}, \cos t = \frac{\sqrt{2}}{2}, \tan t = 1, \cot t = 1, \sec t = \sqrt{2}, \csc t = \sqrt{2}$

$$108. \sin t = \frac{1}{2}, \cos t = \frac{\sqrt{3}}{2}, \tan t = \frac{\sqrt{3}}{3}, \cot t = \sqrt{3}, \sec t = \frac{2\sqrt{3}}{3}, \csc t = 2$$

$$109. \sin t = -\frac{\sqrt{3}}{2}, \cos t = -\frac{1}{2}, \tan t = \sqrt{3}, \cot t = \frac{\sqrt{3}}{3}, \sec t = -2, \csc t = -\frac{2\sqrt{3}}{3}$$

$$110. 1.015$$

$$111. -0.228$$

$$112. 1.051$$

$$113. -2.414$$

$$114. -1.414$$

$$115. 1.414$$

$$116. -7.115$$

$$117. 1.540$$

$$118. -1.192$$

$$119. 1.556$$